

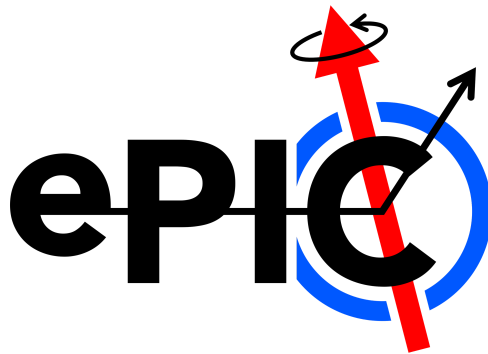
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LightGBM-Guided Exploration of a Six-Dimensional ePIC nHCal Design Space for Low-Energy Neutron Detection

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Abstract

The ePIC negative-eta hadronic calorimeter (nHCal) is a backward sampling calorimeter intended to cover hadron detection in the electron-going region, including neutral-hadron detection and related hit-level activity measurements. This note tests whether the simulated low-energy neutron hit-detection efficiency can be improved by varying the absorber and scintillator thicknesses in three longitudinal nHCal segments. These front, middle, and back segments provide a coarse description of the shower start, middle, and tail, forming a six-dimensional geometry design space. Candidate geometries were generated with a standalone DD4hep/DDsim workflow and simulated with Geant4 using a neutron G4GPS source spanning 0.1–2.5 GeV/c. A LightGBM surrogate-guided search was used to rank geometries using neutron detection efficiency, average fired-tile count, and average fired-layer count. The sampled landscape was shallow: the final selected geometry produced a higher efficiency point estimate than the baseline, increasing from 0.53168 ± 0.00223 to 0.53666 ± 0.00223 , and fired-tile and fired-layer activity increased by about 1%. These results show that surrogate-guided optimization provides a practical way to explore the nHCal geometry design space and identify geometry-dependent changes in neutron hit-detection efficiency. In this scan, the spectrum-integrated gain is about 0.94%, corresponding to about 1.6 combined standard errors. The largest binned improvement appears in the first kinetic-energy bin at about 60%, with broader positive differences across the intermediate neutron-energy region. Additional higher-statistics and systematic studies can further map the design space.

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1 Introduction

The Electron-Ion Collider will use the ePIC detector to study the structure of nucleons and nuclei through high-luminosity electron-proton and electron-ion collisions. In this environment, forward and backward calorimetry provide important information about neutral particles and calorimeter hit activity outside the central tracking region. The negative-eta hadronic calorimeter (nHCal) is a backward sampling calorimeter intended to cover hadron detection in the electron-going region. With neutral hadrons being difficult to detect, optimizing the response to low-energy neutrons is therefore relevant for understanding the hit-level performance of the backward hadronic calorimetry region.

Within that detector context, hadronic calorimeters are typically designed under a limited set of geometric and material constraints, and previous optimization studies often focus on choices such as total layer count and absorber-to-scintillator ratio. In these studies, the sampling structure is commonly treated as uniform across depth. This motivates a natural question: whether allowing the absorber and scintillator thicknesses to vary non-uniformly can improve particle detection efficiency.

This note studies that question for low-energy neutron detection at the nHCal. The reference design is a uniform 10-layer sampling calorimeter with identical absorber and scintillator thickness in every segment. The alternative explored here is a nonuniform longitudinal parameterization in which the calorimeter is divided into three segments, each with its own absorber and scintillator thickness. The three segments correspond to a coarse front, middle, and back description of the shower start, middle, and tail while avoiding the less realistic case in which every layer is independently optimized.

Because the resulting parameter space is continuous and multi-dimensional, a brute-force scan is impractical. The optimization is thus approached with a surrogate-guided workflow: simulated detector responses are used to train a LightGBM regression model, and that model is then used to guide subsequent proposals of high-performing candidates.

This study uses that workflow to test whether a small number of full simulations can identify nonuniform nHCal geometries with improved low-energy neutron hit detection, and to measure how large that improvement is relative to the uniform baseline.

2 Analysis details

This section summarizes the event generation, detector response, signal extraction, surrogate-guided proposal loop, and final comparison used in this study.

2.1 Data and Event Generation

This study used simulated neutron events in a simplified standalone model of the ePIC backward/negative hadronic calorimeter (nHCal). No collision data were used. Geometry descriptions were generated from a parameterized DD4hep compact XML template in which the calorimeter was modeled as a 10-layer sampling detector with a 3/3/4 longitudinal segmentation pattern. The geometry generation and simulation workflow was run as a standalone DD4hep/DDsim study outside the full ePIC detector simulation framework to simplify large-scale candidate generation and analysis.

The transverse detector area was fixed to $1\text{ m} \times 1\text{ m}$ with $10\text{ cm} \times 10\text{ cm}$ readout cells.

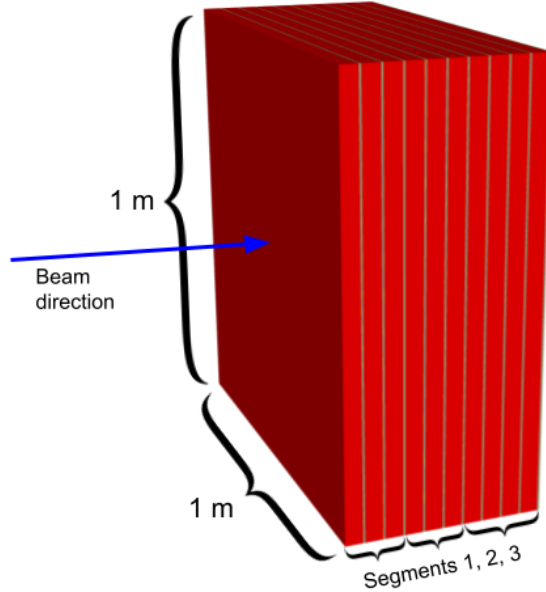


Figure 1: *Detector sketch showing the three longitudinal segments and neutron beam direction used in the study. Red regions indicate absorber material, thin yellow bands indicate scintillator layers, and thin gray bands indicate air gaps. The scintillator and air-gap layers are much thinner than the absorber plates and therefore may be hard to distinguish in the sketch.*

The detector response was simulated with DD4hep/DDsim 1.36.0, Geant4 11.4.1, and ROOT 6.38.04 [1–5]. The geometry parameterization, analysis scripts, and surrogate workflow were run in Python using NumPy, SciPy, scikit-learn, and LightGBM. Neutron events were simulated via a neutron G4GPS beam configuration spanning a continuous momentum range of 0.1–2.5 GeV/ c . The neutron gun was a point source at the origin directed toward the nHCal. For neutrons, this momentum range corresponds to kinetic energies from about 5 MeV to 1.7 GeV. To model the low-energy portion of the spectrum with more accurate neutron cross sections, the samples used the Geant4 high-precision neutron physics list `FTFP_BERT_HP`, passed through DDsim. The tabulated source rises to a maximum near 0.3 GeV/ c and then falls approximately exponentially, providing a DIS-inspired neutron spectrum for this study, with linear interpolation between points performed by G4GPS. The DDsim runs used Geant4 particle-gun mode with G4GPS enabled, kept the full generated-event record, disabled the particle-history cuts by setting the minimum kinetic-energy and minimum parent-vertex-distance cuts to zero, and set the random seed explicitly for each run.

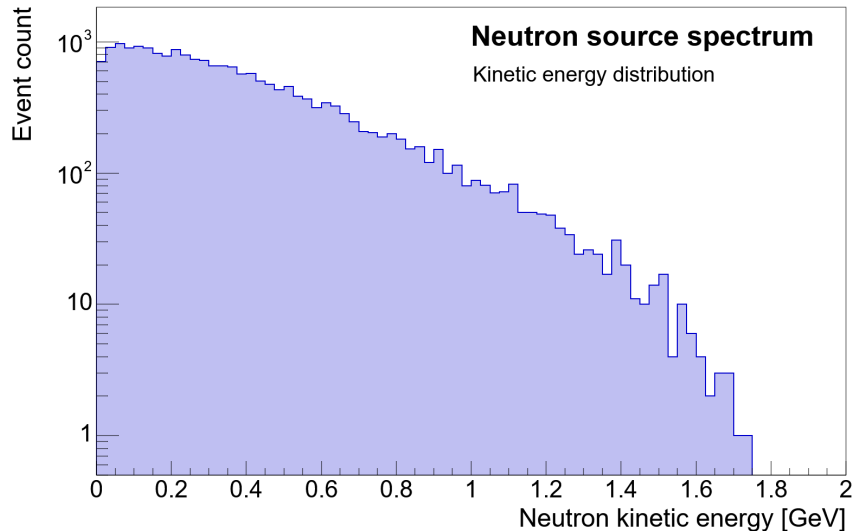


Figure 2: *Simulated neutron source spectrum used in the DDsim study, shown as a function of neutron kinetic energy. The distribution comes from the tabulated G4GPS input histogram with linear interpolation between points.*

The baseline used steel absorber plates and polystyrene scintillator with uniform nHCal thicknesses, where each layer has a 4.0 cm absorber plate, 0.05 cm air gap, 0.4 cm scintillator layer, and 0.05 cm air gap. The air-gap thickness was kept fixed in all geometries. Candidate geometries were defined by allowing the absorber and scintillator thickness to vary independently in each of the three longitudinal segments. The absorber thickness in each segment was allowed to vary from 3.5 to 4.5 cm, and the scintillator thickness in each segment was allowed to vary from 0.3 to 0.6 cm. These per-segment bounds limited the possible overall detector depth to approximately -13% to $+16\%$ relative to the uniform baseline, without a separate total-depth constraint.

The initial exploration used a 150-point Latin-hypercube sample, a space-filling sampling method that spreads points across the allowed parameter ranges, to broadly cover the six-dimensional design space, with 5,000 simulated events per geometry. These values were chosen as a practical compromise between design-space coverage and simulation cost for the first pass. Together, the initial scan, surrogate-guided iterations, higher-statistics checks, and final matched comparisons used approximately 1.3 million simulated neutron events.

2.2 Signal Extraction and Efficiency Definition

Thresholds were derived from a minimum-ionizing-particle (MIP)-based calibration step. A muon calibration script was used to extract a reference MIP scale from the baseline geometry. The calibration used 10,000 10 GeV negative-muon events from a point source at the origin directed toward the nHCal. Energy deposits were first summed by layer for each event and then pooled by longitudinal segment to extract three fitted MIP values. Each segment distribution was histogrammed in 400 bins from zero to an upper edge set to the larger of 1.15 times its 99.5th percentile and six times its median. A ROOT Landau function was fit from 0.25 times the histogram peak-bin center to three times that value, capped at the histogram upper edge. The fitted Landau location and width parameters, μ and σ , were converted to the distribution mode using $\text{MPV} = \mu - 0.22278298\sigma$ [6]; the histogram peak-bin center was used if the fit failed. The three segment MPVs were 0.6184, 0.6201, and 0.6194 MeV and were averaged into a single 0.6193 MeV reference MPV for the 4 mm baseline scintillator thickness. Segment-specific thresholds for a given geometry were then obtained by linearly scaling this reference MPV by the segment scintillator thickness and multiplying by the chosen MIP fraction parameter. In the optimization runs reported here, the threshold setting used a MIP fraction of 0.5. Figure 3 shows a representative example: the segment-1 baseline muon layer-energy distribution from the 10,000-event muon calibration sample, fit with a Landau function over the main MIP peak region to extract one of the three MPVs used to form the averaged reference.

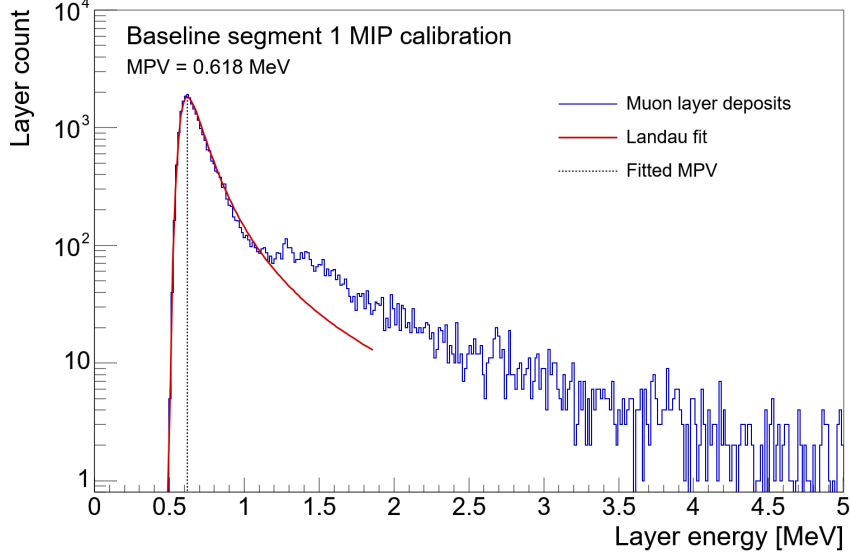


Figure 3: *Muon layer-energy distribution for segment 1 of the baseline geometry. The Landau fit is restricted to the main MIP peak region and is used to extract one of the fitted MPVs used to form the averaged threshold-calibration reference.*

The processed neutron samples used a 0–100 ns integration window to sum hit contributions into each cell energy. A cell was counted as fired if this integrated energy was greater than or equal to the threshold assigned to the corresponding longitudinal segment. A layer was counted as fired if at least one cell in that layer fired. This thresholding procedure defines a hit-activity metric rather than a full neutron-reconstruction efficiency. This MIP-based thresholding approach follows the same logic used in a previous nHCal study [7].

All generated neutron events with a positive recorded generator energy were included in the efficiency denominator. An event was counted as detected if at least one calorimeter cell met or exceeded its threshold anywhere in the detector. The main performance metric, detection efficiency, was then defined as

$$\epsilon_{\text{det}} = \frac{N_{\text{detected}}}{N_{\text{valid}}}.$$

In addition to detection efficiency, two shower-activity observables were recorded: the average number of fired tiles per valid event and the average number of fired layers per valid event. A fired tile is a readout cell with energy greater than or equal to the segment threshold, and a fired layer is a calorimeter layer containing at least one fired tile.

2.3 Detector-Level Scope and Containment

This study focuses on hit-level detector response from simulated calorimeter energy deposits. Reconstruction-level effects such as clustering, energy reconstruction, and particle identification are not applied, since the goal is to compare geometry-dependent hit detection under a fixed surrogate-guided workflow. The same beam configuration, detector template, threshold definition, and hit-processing logic were used for the baseline and candidate geometries, so observed differences are interpreted as geometry-dependent responses within this fixed setup.

As a transverse-containment check, the visible energy deposited in the outermost transverse cell ring, summed across all layers, was compared with the total visible HCal energy. In the checked samples, the outer ring contained less than 2.5% of the visible energy. Since this ring corresponds to the outer 10 cm of the 1 m by 1 m detector area and covers a large fraction of the transverse area, this fraction provides a transverse-containment proxy and indicates that edge activity is small.

2.4 Surrogate Model and Candidate Proposal

The optimization stage used a surrogate-guided loop. Simulated event outputs were processed and converted into a geometry-level table with one row per geometry. Each row contained the absorber and scintillator thicknesses together with averaged performance metrics. The candidate geometry proposal loop used the composite score described below to select candidates for simulation. The final selected geometry was then evaluated against the baseline with matched high-statistics runs, using neutron detection efficiency as the primary comparison metric. The surrogate input features were:

- absorber thicknesses in segments 1, 2, and 3, each bounded from 3.5 to 4.5 cm
- scintillator thicknesses in segments 1, 2, and 3, each bounded from 0.3 to 0.6 cm

Together, these six thicknesses define the design space explored in this study. The surrogate was used only within these predefined bounds, so candidate selection did not rely on extrapolation outside the scanned design space. The target variables were:

- neutron detection efficiency
- average fired-tile count
- average fired-layer count

The regression model was implemented with LightGBM [8]. The prediction for each target

can be expressed via

$$\hat{y}(x) = \sum_{m=1}^M \eta f_m(x),$$

where x is the vector of geometry inputs, M is the number of boosted trees, η is the learning rate, and f_m is the prediction from tree m . The multi-output setup wrapped three separate LightGBM regressors, one for each target observable. The current configuration used 500 estimators, a learning rate of 0.05, 31 leaves, and a feature-subsampling fraction of 0.8. As a model check, 20% of the final geometry-level table was held out for validation. The mean absolute percentage errors were 1.04% for efficiency, 1.57% for fired-tile count, and 1.63% for fired-layer count. These hyperparameters were fixed settings and were not separately tuned in this study.

The surrogate ranking used a normalized weighted score built from predicted neutron detection efficiency, average fired-layer count, and average fired-tile count, with weights of 0.5, 0.3, and 0.2, respectively. For each metric, the normalization used the baseline geometry as the zero-point anchor and the 90th percentile of the initial Latin-hypercube sample as the 100-point reference scale. Anchoring at the baseline expressed the score as improvement over the reference design, while assigning no credit below baseline prevented underperformance in one metric from offsetting gains in another. The 90th percentile provided a representative high-performance scale without allowing a noisy maximum to compress the remaining score range. Values between the baseline and the 90th percentile were scaled linearly. Values above the 90th percentile were allowed to saturate at the upper anchor during surrogate candidate ranking so that an extreme prediction in one metric could not dominate the intended weights. When the observed geometry table was rescored directly, values above the 90th percentile continued increasing with a reduced slope of 0.2, retaining discrimination among strong observed geometries while limiting sensitivity to upper-tail fluctuations.

These weights prioritized neutron detection efficiency as the primary objective, and the larger secondary weight on fired-layer multiplicity gave extra preference to longitudinal development in a study focused on longitudinal segmentation. The smaller fired-tile weight was retained as a weaker measure of overall activity.

For later iterations, each proposal step rescored a pool of 20,000 geometries generated within the predefined design bounds using scrambled Sobol sampling, which provided broad, even coverage of the six-dimensional design space with fewer clusters and gaps than ordinary random sampling. The same seed was reused, so the candidate pool remained fixed across iterations while the retrained surrogate predictions changed. The candidates were converted into surrogate feature rows, their performance was estimated, and 5 geometries were selected per batch. A minimum normalized geometry-space separation of 0.05 was enforced between candidates selected within each batch as a practical diversity setting. The pool size and batch size were chosen to balance broad surrogate-guided exploration with downstream simulation cost.

After each batch of geometries was simulated and processed, the geometry-level table was updated and used to retrain the surrogate model. The updated model was then used to propose the next batch of promising candidates. In parallel with this loop, the geometry-level table was rescored directly using the same metric weights and normalization anchors to identify the current best observed geometry.

2.5 Statistical and Systematic Uncertainties

The main uncertainty treated in this study was Monte Carlo statistical variation between geometry evaluations. This mattered because a shallow efficiency landscape can cause low-statistics evaluations to mis-rank near-top candidates even when the broader performance region is identified correctly.

The surrogate score ranked point predictions without an explicit prediction-uncertainty term, leaving candidate ordering sensitive to statistical fluctuations in the training targets. Surrogate smoothing provided partial mitigation, while the higher-statistics reruns supplied the main validation of promising candidates.

To address this, the study used a staged event-count strategy. High-scoring candidates from the initial 5,000-event scan were evaluated across four independent 5,000-event runs and promoted through matched multi-seed comparisons before final selection as a method to reduce the influence of low-statistics fluctuations on candidate choice.

For the final baseline-versus-selected comparison, both the baseline geometry and the selected geometry were evaluated with 50,000 events. The quoted efficiency uncertainties are binomial standard errors computed from the detected and valid event counts, and the fired-tile and fired-layer uncertainties are standard errors on the mean computed from the event-by-event spreads in the same samples. For analyses of individual kinetic-energy bins, the plotted efficiency uncertainties are likewise binomial standard errors computed independently in each bin from the detected and valid event counts.

A broader treatment of systematic variations, such as alternative threshold settings, physics lists, or beam-shape variations, is outside the scope of the present note. The linear scaling of the MIP threshold with scintillator thickness is also treated as part of the fixed workflow and was not varied separately.

2.6 Results and Discussion

Tables 1–4 summarize the baseline, the selected geometry, and two additional geometries included in the higher-statistics checks. The selected nonuniform geometry produced a higher neutron detection-efficiency point estimate than the baseline, with the broadest positive pattern appearing in the intermediate neutron-energy region and a less statistically distinguishable spectrum-integrated increase in the final matched comparison. Across three surrogate-guided iterations, the proposed candidates repeatedly clustered in a similar region of geometry space. Quantitatively, all 15 surrogate-proposed candidate slots had thinner-than-baseline front absorbers, 14/15 had thicker middle scintillator, 15/15 had thicker back scintillator, and 12/15 had thicker back absorber. This recurrence indicates a stable surrogate preference within the fixed candidate pool; establishing whether this region is unique would require broader validation. Because the later proposals continued to revisit this same longitudinal pattern without producing a clearly stronger validated candidate, the scan was stopped after the third iteration and the final selected geometry was taken from the higher-statistics checks in this region. In the final matched comparison, the neutron detection efficiency increased from 0.53168 ± 0.00223 (Table 1) to 0.53666 ± 0.00223 (Table 2), corresponding to an absolute gain of 0.00498 and a relative gain of about 0.94%.

All absorber and scintillator thicknesses are given in cm and rounded to four decimal places. Efficiency and shower-activity metrics are rounded to five decimal places. The baseline has a total absorber depth of 40.00 cm and a total scintillator depth of 4.00 cm, and the selected geometry has a total absorber depth of about 40.43 cm and a total scintillator depth of about 4.32 cm. Using iron as an approximate proxy for the stainless-steel absorber and polystyrene for the scintillator, and neglecting the fixed air gaps, the effective nuclear interaction depth was estimated as

$$N_{\lambda_I} = \frac{t_{\text{abs}}}{\lambda_{I,\text{Fe}}} + \frac{t_{\text{scin}}}{\lambda_{I,\text{PS}}},$$

with $\lambda_{I,\text{Fe}} = 16.77$ cm and $\lambda_{I,\text{PS}} = 77.07$ cm from the PDG material tables [9, 10]. This gives about 2.44 and 2.47 nuclear interaction lengths for the baseline and selected geometry, respectively.

Candidate A (Table 3) and candidate B (Table 4) each summarize the mean performance from four independent 5,000-event runs. The tables report their mean values without quoted uncertainties.

Metric	Value
$t_{\text{abs},1}$	4.0000
$t_{\text{scin},1}$	0.4000
$t_{\text{abs},2}$	4.0000
$t_{\text{scin},2}$	0.4000
$t_{\text{abs},3}$	4.0000
$t_{\text{scin},3}$	0.4000
Efficiency	0.53168 ± 0.00223
Tiles Mean	1.09176 ± 0.00659
Layers Mean	0.97714 ± 0.00566

Table 1: *Baseline geometry.*

Metric	Value
$t_{\text{abs},1}$	3.5894
$t_{\text{scin},1}$	0.3793
$t_{\text{abs},2}$	3.9174
$t_{\text{scin},2}$	0.4802
$t_{\text{abs},3}$	4.4785
$t_{\text{scin},3}$	0.4348
Efficiency	0.53666 ± 0.00223
Tiles Mean	1.10520 ± 0.00661
Layers Mean	0.98940 ± 0.00569

Table 2: *Selected geometry.*

Metric	Value
$t_{\text{abs},1}$	3.6668
$t_{\text{scin},1}$	0.3223
$t_{\text{abs},2}$	4.0215
$t_{\text{scin},2}$	0.5642
$t_{\text{abs},3}$	4.4091
$t_{\text{scin},3}$	0.5773
Efficiency	0.53465
Tiles Mean	1.08775
Layers Mean	0.97765

Table 3: *First higher-statistics check candidate.*

Metric	Value
$t_{\text{abs},1}$	3.6095
$t_{\text{scin},1}$	0.3909
$t_{\text{abs},2}$	3.5670
$t_{\text{scin},2}$	0.5264
$t_{\text{abs},3}$	4.4344
$t_{\text{scin},3}$	0.4925
Efficiency	0.53405
Tiles Mean	1.11270
Layers Mean	0.99910

Table 4: *Second higher-statistics check candidate.*

Using the 50,000-event baseline and selected-geometry samples, the efficiency difference corresponds to about 1.6 combined standard errors. Over the same comparison, the average fired-tile multiplicity increased from 1.09176 ± 0.00659 to 1.10520 ± 0.00661 , and the average fired-layer multiplicity increased from 0.97714 ± 0.00566 to 0.98940 ± 0.00569 . These correspond to increases of about 1.23% and 1.25%, respectively.

Figure 4 compares the baseline and selected-geometry efficiencies as a function of neutron kinetic energy to check how the integrated result depends on the generated spectrum. The plot uses the 50,000-event baseline and selected-geometry neutron samples, applies the geometry-specific 0.5-MIP thresholds during event processing, groups events into the fixed kinetic-energy bins used by the plotting macro, and draws binomial error bars from the detected and valid event counts in each bin.

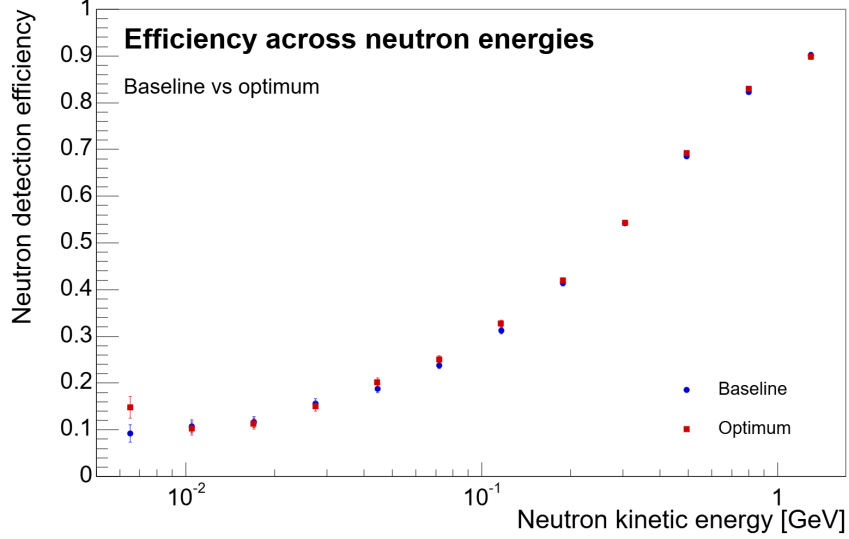


Figure 4: *Neutron detection efficiency as a function of neutron kinetic energy for the baseline and selected geometries. The plot was produced from the 50,000-event neutron samples for each geometry using the nominal 0.5-MIP threshold, fixed kinetic-energy bins, and binomial error bars.*

The largest improvement occurs in the first kinetic-energy bin, where the relative efficiency gain is about 60%; this bin contains about 0.48% of the generated neutrons. Most of the remaining positive differences occur across the intermediate region, roughly 0.034–0.233 GeV, which contains about 35.4% of the generated neutrons; across these bins, the selected geometry exceeds the baseline by roughly 1.5–7.5% in relative terms, with an average binned gain of about 4.7%. Despite overlapping binomial uncertainties in individual bins, the consistently positive intermediate-bin differences are compatible with a geometry-dependent trend. The overall spectrum-integrated efficiency gain is about 0.94%.

Figure 5 compares the same baseline and selected-geometry samples as a function of MIP threshold fraction to check how the result depends on the hit threshold. The scan was produced by reprocessing the same two 50,000-event neutron samples over a sequence of threshold fractions using each geometry’s own calibration constants. Because the same events were reused, points along each curve are correlated across threshold values.

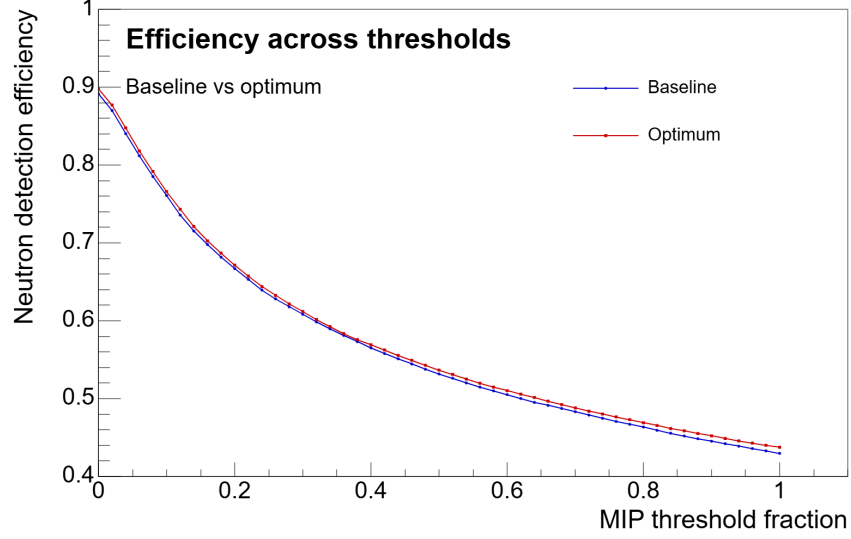
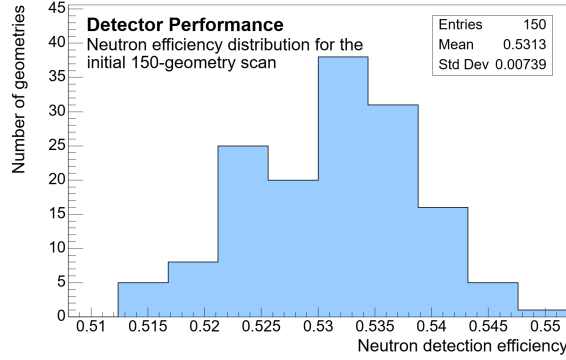
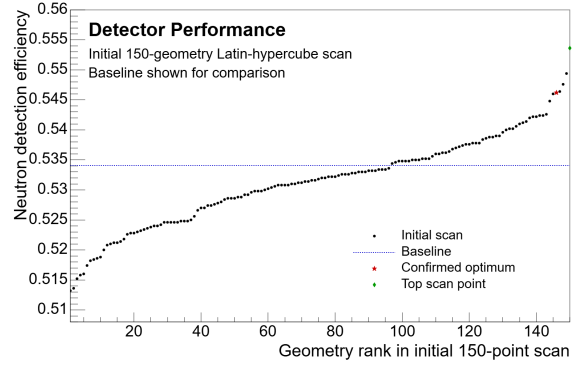


Figure 5: *Threshold-dependence check for the baseline and selected geometries across the scanned MIP-threshold range.*

The broader optimization landscape is summarized in Figure 6. Both panels are derived from the initial 150-geometry survey of the design space, with one neutron-efficiency value per geometry from the 5,000-event evaluations; the left panel shows the efficiency distribution, and the right panel shows the same values sorted by efficiency with the baseline overlaid for reference.



(a) *Histogram of neutron detection efficiencies across the initial 150-geometry scan.*



(b) *Sorted neutron detection efficiencies from the initial 150-geometry scan, with the baseline shown for comparison.*

Figure 6: *Neutron detection efficiency views for the initial 150-geometry scan.*

The initial 150-point scan covered a narrow efficiency range, approximately 0.513 to 0.554, broadly consistent with nHCal performance estimates [11]. This indicates a shallow response landscape in which many sampled geometries were close in performance, with the baseline

near the center of the sampled range. In such a landscape, near-top candidates are harder to distinguish with limited event statistics because Monte Carlo fluctuations can be comparable to the true geometry-to-geometry differences.

3 Conclusion

This study tested whether nonuniform longitudinal nHCal thicknesses could improve low-energy neutron hit detection relative to a uniform baseline within a fixed simulation and surrogate-guided workflow. The sampled geometries produced a shallow response landscape, with many candidates close to the baseline and to one another.

The selected nonuniform geometry produced higher neutron detection efficiency and fired-tile and fired-layer point estimates than the uniform baseline, while other checked geometries remained comparable within the available statistics. Across three surrogate-guided iterations, the proposed candidates repeatedly clustered in a similar region of the fixed candidate pool, and the final selected geometry was taken from this region after higher-statistics checks. In the final 50,000-event baseline-versus-selected comparison, the efficiency increased from 0.53168 ± 0.00223 to 0.53666 ± 0.00223 , corresponding to a spectrum-integrated gain of about 0.94% and about 1.6 combined standard errors. The largest binned improvement appears in the first kinetic-energy bin at about 60%, and the broader positive differences across the intermediate neutron-energy region are compatible with a geometry-dependent trend.

The surrogate model provided a practical way to reuse the simulated geometry sample when choosing later candidates. Approximately 1.3 million simulated neutron events were used to identify the shallow structure of the sampled response landscape, and near-tied geometries still required higher-statistics checks to assess small efficiency differences. Because the surrogate ranked point predictions without an explicit uncertainty term, these checks were central to the final candidate selection. Future studies can expand the design space by allowing absorber and scintillator thicknesses to vary over wider ranges, together with larger samples, broader systematic variations, and extensions to other neutral hadrons.

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